

Monitoring System for Efficient ESS Charging/Discharging Scheduling based on Intelligent IoT Platform

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Abstract

In the case of charging/discharging of the existing Energy Storage System (ESS), since the time is predetermined, the optimal time zone that changes depending on the external environment and characteristics of the building cannot be applied, resulting in wasted energy.

In this study, we construct a database (DB) by collecting internal and external environmental data and real-time space usage data through IoT for building and analyze the patterns of power peak times. In addition, in order to efficiently use the energy stored in the ESS, charging is performed at a time when power usage is low, discharging is performed at a time when power usage is rapidly increased, and it is visually managed through a monitoring system.

Through this, it is expected that energy efficiency may be increased, energy independence rate may be increased, and ultimately carbon reduction and carbon neutrality may be achieved by reducing the amount of wasted power.

Keywords: Intelligent IoT, Energy consumption patterns, ESS charging/discharging scheduling.

1. Introduction

Global demand resource markets and efficient energy management, renewable energy generation, and energy consumption pattern analysis are important for responding to climate change and energy shortages. In particular, the convergence of technologies such as IoT, artificial intelligence, and big data is accelerated in the 4th industrial revolution, and it is necessary to intelligence the energy platform. In addition, in the case of the current Energy Storage System (ESS), the charging/discharging time is fixed, and wasted energy is generated, and in order to reduce it and maximize energy efficiency, ESS charging/discharging should be analyzed to stably

reduce operation and cost. The existing method advances through connection with consumption patterns and photovoltaic power generation, and thus the optimal ESS charge/discharge scheduling is required. Therefore, it is necessary to develop and research energy supply processes according to energy data consumption patterns by applying ESS to building units based on IoT and artificial intelligence.

In addition, future industries will expand around data, and Autonomous IoT and Intelligent IoT will be the center of the industry. Thus, to counteract this, we serve users by simplifying the sharing mechanisms of data collected through smart devices and sensors from the perspective of Data-Sharing Made Easy and The Tech Stack goes Physical in Deloitte's Tech Trends 2022. And according to Gartner's Top Strategic Technology Trends for 2022, it is important to monitor and manage data from a Data Fabric perspective, regardless of where it is stored. As such, the importance of data and sharing is increasing.

Focusing on this data, according to Forbes' The 5 Biggest Technology Trends In 2022, more investment will be made in energy generation through distributed power over the next few years. And at the Consumer Electronics Show (CES) in 2022, we proposed a service that can optimize energy and detect new and renewable energy to supply energy to use at the most optimal time of the day.

Based on the intelligent IoT platform we propose, we provide services through monitoring by linking data analysis, intelligent IoT, and new and renewable energy for efficient ESS charging/discharging scheduling according to energy consumption patterns. If the linked system is applied to various industries, ESS can be operated efficiently, cost can be reduced, and carbon neutrality can be realized through carbon reduction.

2. Related Works

Emphasizing the framework for intelligent energy management and the need for energy data management, analysis and sharing, it ultimately proposed an energy business model [1]. We proposed a framework for monitoring the consumption and indoor condition data of buildings by integrating IoT sensors [2]. It proposed an IoT energy platform for managing energy data in buildings [3]. If the grid has renewable resources for energy production and ESS, the proposed technology has verified operational costs and scheduling accuracy [4]. Therefore, this author proposes a monitoring method by explaining the preceding research of efficient ESS charging/discharging scheduling according to energy consumption patterns based on intelligent IoT platform and analyzing the differentiation.

3. ESS Charging/Discharging Scheduling Architecture based on Intelligent IoT Platform

The association between energy data and charge/discharge time is analyzed to derive optimal ESS charge/discharge scheduling.

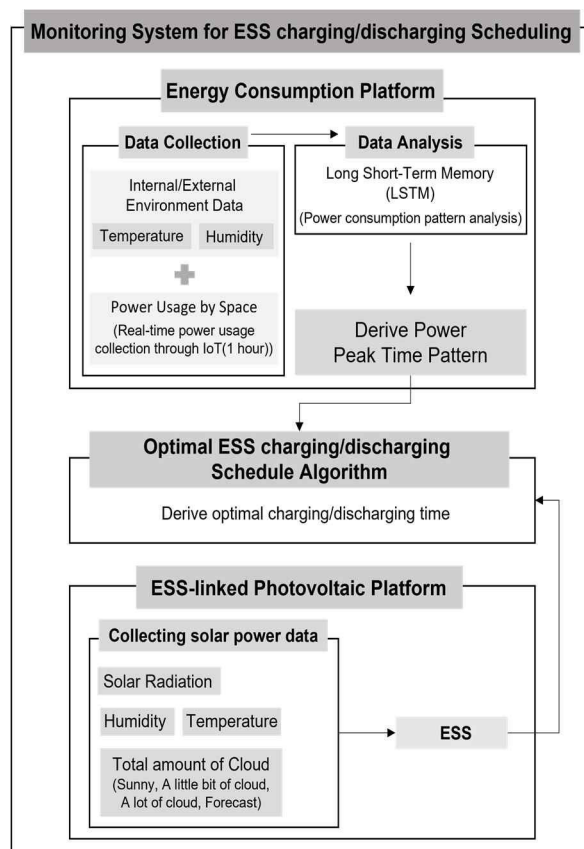


Fig 1 : ESS charging/discharging Scheduling Architecture

Fig 1 illustrates the proposed architecture, which derives the optimal ESS charge/discharge scheduling and presents the monitoring system that manages it.

3.1 Energy Consumption Platform based on Intelligent IoT

To derive optimal ESS charging/discharging scheduling for buildings, data on the internal and external environments of the building are collected first. The external environment data sets temperature, humidity, and two environmental variables through weather forecasts provided by the Korea Meteorological Administration. In the case of internal environment data, the data of power usage by space is collected in real time through IoT installed inside the building. It analyzes the consumption pattern through the Long Short-Term Memory (LSTM) algorithm and identifies the pattern in the time zone with the highest energy usage. Accordingly, we present an intelligent IoT-based energy consumption platform that analyzes energy consumption patterns and data for each space and operates intelligently.

3.2 ESS-linked Photovoltaic Platform

ESS charging/discharging technology linked to renewable energy (solar light) is applied for efficient use of electricity for each building. In the proposed platform, environmental variables that affect solar energy (PV) power generation were set as temperature, humidity, solar radiation, and total cloud. Electric cloud is classified as 'clear', 'a little cloud', 'cloudy', and 'cloudy', and changes in these variables affect ESS charge/discharge.

The optimal ESS charge/discharge scheduling is derived by linking two platforms to supplement the intermittent problem of new and renewable energy that is heavily influenced by external environmental factors and to reduce energy waste using the intermittent problem.

3.3 ESS Charging/Discharging Scheduling for Energy Demand Management

The optimal ESS charge/discharge scheduling algorithm is derived by analyzing the connection between the power peak time in building and ESS charge/discharge technology derived through the energy consumption platform, and finally, a plan to monitor and manage it is presented. The power peak time is predicted according to seasons and building characteristics, and the optimal charging/discharging time that can move flexibly depending on the power peak time is derived from the existing charging/discharging method. Since the time is set

fluidly, wasted energy may be reduced, and the energy independence rate may be increased by being grafted according to each characteristic.

4. Implementation of Monitoring System

A monitoring system that can manage the optimal ESS charging/discharging scheduling according to energy consumption pattern analysis and renewable energy-linked ESS is presented. Through this, it quickly detects the state of the system, responds quickly and accurately to problem situations, and ultimately manages energy efficiently.

4.1 Analysis of Energy Consumption Patterns

The energy consumption pattern is analyzed considering that the power peak time will vary depending on the characteristics of each building.

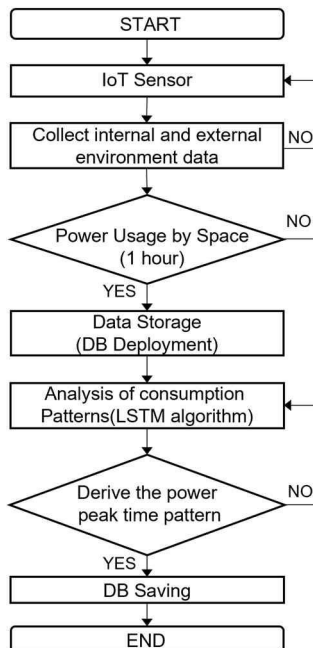


Fig 2 : Energy Consumption Pattern Analysis Algorithm

Fig 2 is a proposed algorithm, which builds internal/external environmental data into a database and uses the LSTM algorithm to identify the pattern of power peak times.

4.2 ESS Charge/Discharge Scheduling Algorithm for Optimal Operation

In the case of the existing ESS charging/discharging, since the charging/discharging time is predetermined, wasted energy is generated and energy efficiency is reduced. Therefore, this research proposes an optimal ESS charge/discharge scheduling algorithm that could maximize energy

efficiency and increase energy independence by identifying seasonal and building characteristics and identifying power peak time patterns that vary accordingly.

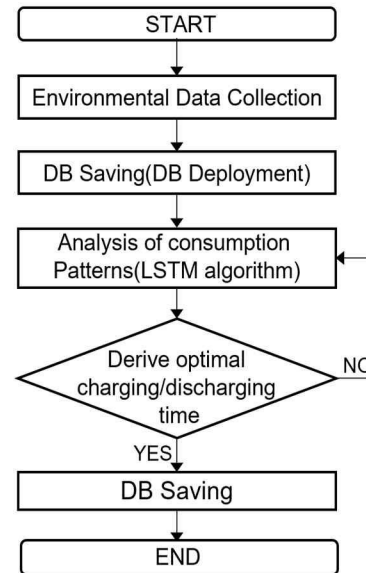


Fig 3 : ESS charging/discharging scheduling algorithm

Fig 3 illustrates the proposed algorithm, which derives the optimal ESS charging/discharging time through the DB and LSTM algorithms.

4.3 Implementation of ESS Charge/Discharge Scheduling Monitoring System

In this work, we present a monitoring system that can efficiently manage two platforms.



Fig 4 : ESS charge/discharge Scheduling Monitoring System

Fig 4 is the proposed monitoring system. First, it presents the accuracy of energy consumption pattern analysis. As shown in the figure, it shows the accuracy in percentage and shows how much the building's power consumption has changed compared to the previous month. In addition, the analysis of consumption patterns by building is

shown in a graph, and the x-axis represents time and the y-axis represents the amount of power. The time was from 0:00 to 23:00, and the amount of power was from 0 kWh to a maximum of 700 kWh. Power consumption usually increased from 8:00. and then decreased from 17:00

An optimal charging/discharging time was derived based on the power consumption pattern. The charge and discharge of the ESS are visualized in the figure, and the charge and discharge times are represented in graphs. The charging and discharging were performed every hour where the time for the most charging and the time for the most power consumption overlap.

It also shows the ESS storage amount and the energy independence rate when optimal charge/discharge scheduling is derived. Finally, a data table used for analysis was presented.

Through this monitoring system, it is expected that energy can be efficiently managed to reduce carbon and increase the self-reliance rate of new and renewable energy.

5. Result

This research derives the optimal ESS charge/discharge scheduling algorithm by linking an energy consumption platform with an ESS-linked photovoltaic power generation platform and efficiently manages the algorithm through implementation of a monitoring system for ESS charge/discharge scheduling. It collects internal/external environmental data and data on power usage by space collected on an hourly basis through IoT and analyzes them to identify patterns of power peak times.

In the past, if ESS charge/discharge technology could not flexibly respond to power peak times that change depending on the external environment and characteristics of each building, the power peak time pattern analysis should charge when power is low and discharge when power is high. Accordingly, it is expected to reduce wasted energy, increase energy independence, and ultimately contribute to carbon neutrality.

Acknowledgment

This work was supported by the Human Resources Development (No.20204030200090) of the Korea Institute of Energy Technology Evaluation and Planning(KETEP) grant funded by the Korea government Ministry of Trade, Industry and Energy, and This work was supported by the Korea Institute of Energy Technology Evaluation and Planning(KETEP) and the Ministry of Trade, Industry & Energy(MOTIE) of the Republic of Korea (No. 20192710100151), and This research was supported by the MSIT(Ministry of Science and

ICT), Korea, under the ITRC(Information Technology Research Center) support program(IITP-2022-2018-0-01799) supervised by the IITP(Institute for Information & communications Technology Planning & Evaluation)

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